Statistics & Econometrics 2025

Tutorial 2: Problem Set

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1 Multivariate Regression

Exercises to deepen understanding of OLS estimator for multivariate linear regression and corresponding standard errors.

1. After your high quality responses (in the previous tutorial), your manager gives you another task. In light of the upcoming Fed meeting where the new level for the Federal funds rate would be decided, he asks you to finalize a slide deck for a client presentation the next morning that a co-worker had started to work on but couldn't finish for some reason. The presentation was supposed to show how sensitive the returns on your firm's portfolio have been to changes in the federal funds rate in the past. The analysis is based on 13 years of data.

   Luckily, you don't have to compute any numbers as your co-worker had already produced several results. Looking at the codes you see that he had run the following two regressions:

$$\begin{aligned} r_t &= \hat{\alpha} + \hat{\gamma}_1 \Delta i_t + \hat{u}_t \\ &= 5.0 - 2.0 \Delta i_t + \hat{u}_t \\ &= (4.0)(0.5) \end{aligned} \tag{1}$$

   and

$$\begin{aligned} r_t &= \hat{\rho} \Delta i_t + \hat{v}_t \\ &= 0.5 \Delta i_t + \hat{v}_t \\ &= (0.5) \end{aligned} \tag{2}$$

   where the numbers in parentheses represent the standard errors associated with the estimated regression coefficients.

   (a) Question: How do you interpret the two regressions in terms of the sensitivity of your company's portfolio return to changes in the federal funds rate? Test the hypothesis that each of the coefficients equals 0 at the 5% level.

   Answer: The estimated slope coefficient of the first regression (with intercept) implies that the portfolio return will decrease by 2% for a 1% increase in the federal funds rate. The estimated slope coefficient for the second regression (without intercept) implies that your company's return increases by 0.5% for a 1% increase in the fed funds rate.

   Moreover, the slope coefficient in the second regression without an intercept showed a t-statistic equal to $\frac{\hat{\rho}}{SE(\hat{\rho})} = \frac{0.5}{0.5} = 1$, indicating that one cannot reject the null hypothesis the slope coefficient is zero and thus of your company's portfolio return being insensitive to changes in the funds rate. In other words, your firm's portfolio is not exposed to monetary policy risk. This is in contrast to the results of the first regression including an intercept. These suggest that your firm's portfolio return is highly sensitive to changes in the fed funds rate as the slope coefficient $\hat{\gamma}_1$ is highly statistically significant. It's t-statistic is equal to $t_{\gamma_1} = \frac{\hat{\gamma}_1}{SE(\hat{\gamma}_1)} = \frac{-2}{0.5} = -4$ which is below minus the critical value t-crit $= t_{10;1\%} = -3.169$, so it is statistically significant (at least) at the 1% level.

(b) Question: Which of the two regression outputs should you use for the presentation and why?

Answer: You should use the regression that includes an intercept, even though it is not statistically significantly different from zero at conventional significance levels. Running a regression without an intercept results in several problems:

i. The slope estimates on the regressors will be biased.

ii. The regression residuals will not be mean zero.

iii. As a result, the R^2 might be negative.

So even when your intercept is not statistically significant you should include it in the regression.

Assume your boss comes into your office and tells you he has a feeling that your portfolio's return might be more sensitive to variations in inflation and the real federal funds rate (i.e. the nominal funds rate adjusted for the rate of inflation) than simply the nominal federal funds rate alone, but he isn't quite sure. He suggests to simply "let the data speak" and asks you to run a regression of your firm's return on changes in the nominal fed funds rate, changes in inflation, and changes in the real fed funds rate which he suggests to compute as the simple difference between the nominal funds rate and inflation.

(c) Question: How do you respond to this request and why?

Answer: You politely but firmly tell your boss that the suggested regression cannot be run, because the change in the nominal federal funds rate, the change in inflation, and the change in the difference between the two are three perfectly collinear variables. The reason is that the real fed funds rate is simply the difference between the nominal funds rate and the rate of inflation. As a result, the multivariate OLS estimator $\hat{\beta} = (X'X)^{-1}X'y$ and its corresponding variance-covariance matrix $Var(\hat{\beta}) = s^2(X'X)^{-1}$ cannot be computed. The reason is that the matrix $X'X$ will not be of full rank and cannot be inverted.

Let $x_{t1} = \Delta pi_t, x_{t2} = \Delta i_t, x_{t3} = x_{t2} - x_{t1} = \Delta i_t - \Delta \pi_t$

$$X = \begin{bmatrix} x_{11} & x_{12} & x_3 \\ \vdots & \vdots & \vdots \\ x_{T1} & x_{T2} & x_{T3} \end{bmatrix} = \begin{bmatrix} x_{11} & x_{12} & x_{12} - x_{11} \\ \vdots & \vdots & \vdots \\ x_{T1} & x_{T2} & x_{T2} - x_{T1} \end{bmatrix}$$

$$X'X = \begin{bmatrix} x_{11} & \dots & x_{T1} \\ x_{12} & \dots & x_{T2} \\ x_{12} - x_{11} & \dots & x_{T2} - x_{T1} \end{bmatrix} * \begin{bmatrix} x_{11} & x_{12} & x_{12} - x_{11} \\ \vdots & \vdots & \vdots \\ x_{T1} & x_{T2} & x_{T2} - x_{T1} \end{bmatrix}$$
$$= \begin{bmatrix} \sum_t^T x_{t1}x_{t1} & \sum_t^T x_{t1}x_{t2} & \sum_t^T x_{t1}x_{t2} - \sum_t^T x_{t1}x_{t1} \\ \sum_t^T x_{t2}x_{t1} & \sum_t^T x_{t2}x_{t2} & \sum_t^T x_{t2}x_{t2} - \sum_t^T x_{t2}x_{t1} \\ \sum_t^T x_{t2}x_{t1} - \sum_t^T x_{t1}x_{t1} & \sum_t^T x_{t2}x_{t2} - \sum_t^T x_{t1}x_{t2} & \sum_t^T x_{t2}x_{t2} - 2\sum_t^T x_{t2}x_{t1} + \sum_t^T x_{t1}x_{t1} \end{bmatrix}$$

We see that column 3 of X'X is just the difference between column 2 and column 1

$$(X'X)_3 = (X'X)_2 - (X'X)_1$$

Therefore, we're not able to calculate the inverse of X'X and therefore we're not able to calculate the OLS estimator in this case.

Assume your colleague also shows you two other regressions that he ran but didn't include in the presentation. The first was a regression of the change of the unemployment rate on the change of the fed funds rate:

$$\begin{aligned} \Delta unemp_t &= \hat{\alpha_2} + \hat{\gamma}_{12}\Delta i_t + \hat{w}_t \\ &= -0.1 + 0.5\Delta i_t + \hat{w}_t \\ &= \quad (1.1) \quad (0.1) \end{aligned} \tag{3}$$

Moreover, he had estimated a regression of your firm's portfolio return on both the change in the fed funds rate and the unemployment rate:

$$
\begin{aligned}
r_t = & \quad \hat{\alpha} + \hat{\beta}_1 \Delta i_t + \hat{\beta}_2 \Delta unemp_t + \hat{\nu}_t \\
= & \quad 6.0 - 1.0\Delta i_t - 2\Delta unemp_t + \hat{\nu}_t \\
= & \quad (5.0)(0.4) \quad (0.5)
\end{aligned} \tag{4}
$$

where again numbers in parentheses denote the standard errors associated with the estimated coefficients. You have 13 years of data.

(d) Question: Test the hypothesis that both variables, the change in the fed funds rate and the change in the unemployment rate, do not matter for your firm's portfolio at the 5%-level.

What is the null hypothesis in this case? What is the restricted model? If the variance of returns is 25 and you consider 13 years of data, what is the restricted residual sum of squares (RSS)? Further you are given that the RSS of the unrestricted model is 150.

Answer: The null hypothesis is H_0: $\beta_1 = \beta_2 = 0$.
The unrestricted model is $r_t = \alpha + \beta_1 \Delta i_t + \beta_2 \Delta unemp_t + \nu_t$. Under H_0: $\beta_1 = \beta_2 = 0$.

Therefore, the restricted model is $r_t = \alpha + \nu_t$.
To compute the RSS of the restricted model, recall that the sample variance of returns is

$$
\hat{\sigma}^2 = \frac{\sum_{t=1}^{T} (r_t - \bar{r})^2}{T - 1} \tag{5}
$$

which can be rearranged to

$$
\sum_{t=1}^{T} (r_t - \bar{r})^2 = (T - 1) \cdot \hat{\sigma}^2 \tag{6}
$$

Note that for our simple restricted regression on a constant, the residuals are the deviation from the mean $\hat{\nu}_t = r_t - \alpha$

$$
\text{RSS} = \sum_{t=1}^{T} (r_t - \bar{r})^2 = (T - 1) \cdot \hat{\sigma}^2 = 12 \cdot 25 = 300 \tag{7}
$$

The test-statistic for the F-Test is

$$\begin{aligned}
\text{test statistic} &= \frac{RRSS - URSS}{URSS} \times \frac{T-n}{m} \\
\text{where URSS} &= \text{RSS from unrestricted regression} \\
\text{RRSS} &= \text{RSS from restricted regression} \\
m &= \text{number of restrictions} \\
T &= \text{number of observations} \\
n = k+1 &= \#\text{ of regressors in unr. regression incl. constant} \\
&\quad \text{(the total \# of parameters to be estimated)}
\end{aligned}$$

In this question RRRS = 300, URSS = 150 and T-n = 10 and m = 2. Therefore the test-statistic is

$$\text{test statistic} = \frac{300 - 150}{150} \times \frac{10}{2} = 5$$

The critical value is

$$\text{crit} = F_{0.05\%,m,T-n} = F_{0.05\%,2,10} = 4.103$$

As test statistic > crit we reject H_0 $\beta_1 = \beta_2 = 0$ at the 5-% level.

Question: How do you interpret the results from the first regression of the change in the unemployment rate on the change in the funds rate?

Answer: The estimated slope coefficient $\hat{\gamma}_{12} = 0.5$ suggests that a one percent increase in the fed funds rate was associated with a 0.5% increase in the unemployment rate. The associated t-statistic $t_{\gamma_{12}} = \frac{\hat{\gamma}_{12}}{SE(\hat{\gamma}_{12})} = \frac{0.5}{0.1} = 5$ shows that the estimated relationship is strongly statistically significant. Hence, the change in the unemployment rate and the change in the fed funds rate are correlated.

(e) Question: Why is the estimated slope coefficient $\hat{\beta}_1$ on the change in the fed funds rate in the joint regression with the unemployment rate lower than the estimated slope coefficient $\hat{\gamma}_1$ in the univariate regression on only the change in the funds rate?

Answer: The change in the unemployment rate is relevant for explaining variation in the portfolio return as it is highly statistically significant (its t-statistic $t_{\beta_2} = \frac{\hat{\beta}_2}{SE(\hat{\beta}_2)} = \frac{-2}{0.5} = -4$ is far below the 1% critical value). Moreover, it's correlated with the change in the funds rate. Hence, the univariate regression of the portfolio return on the change in the fund rate suffers from omitted variable bias. The bias is given by the expression $\beta_2 \cdot \hat{\gamma}_{12}$ (where β_2 is the true coefficient of which $\hat{\beta}_2$ is an estimate). Since $\hat{\beta}_2$ is highly statistically significant (t-stat of -4), the true β_2 is very likely also negative. This implies that the bias is negative (β_2 is negative and $\hat{\gamma}_{12}$ positive). This means that the estimated coefficient $\hat{\gamma}_1 = -2.0$ in the univariate regression (1) of the portfolio return on the change in the funds rate is too low relative to the true coefficient. It is biased downwards. To properly assess the effect of the change in the funds rate on the portfolio return, one needs to control for the change in the unemployment rate as a relevant correlated variable as done in regression (4).

2 Limited Dependent Variable Models

2. You work in a highly competitive industry where top management is exchanged regularly and your firm has a competitor whose CEOs have been fired quite a few times in the past. Your manager asks you to revise the results of an estimated Probit-Model. The model tries to estimate the probability of your closest competitor's CEO being fired (y_t) in relation to the difference between you and your competitor's sales growth x_t (measured in percent).

$$\begin{aligned} P(y_t = 1|x_t) &= \Phi(\hat{\alpha} + \hat{\beta}x_t) \\ &= \Phi(0.5 + 0.1x_t) \end{aligned}$$

Question: Compute the model-implied probability of $y_t = 1$ for the value $x_t = 5$.

Solution: The model-implied probability is given by

$$P(y_t = 1|x_t) = \Phi(0.5 + 0.1 \cdot 5) = \Phi(1) = 0.84,$$

where $\Phi(1)$ is the CDF of the standard normal distribution evaluated at the point 1. In words, for $x = 5$, the probability that $y = 1$ equals 84 percent.

Question: Compute the marginal effect of a one-unit change in x (starting from $x_t = 5$) on the probability $P(y_t = 1|x_t)$.

Solution: The marginal effect of a one-unit change in x on the model-implied probability is given by

$$\begin{aligned} \frac{dP(y_t = 1|x_t)}{dx} &= \hat{\beta} \cdot f(0.5 + 0.1 \cdot 5) \\ &= 0.1 \cdot f(1) \\ &= 0.1 \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}1^2} \\ &= 0.0242 \end{aligned}$$

where $f(1)$ is the PDF of the standard normal distribution evaluated at the point 1. Recall that the pdf of the standard normal is given by

$$f(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2}$$

In words, the marginal effect of a one-unit change in the explanatory variable x on the probability that $y = 1$ equals 2.42 percent.